Configuration d’un serveur DHCP

Mise en place à faire avant la configuration :

Il faut créer un **conteneur LXC** sur l'hyperviseur Proxmox pour le serveur qui va héberger le service DHCP. Ici, on décide de lui donner comme adresse : 192.168.4.253/24 et on le place dans le VLAN 317 soit le VLAN LAN.

ID ↑	Nom	Pont (br…	Pare-feu	Étiquett…	Adresse MAC	Adresse IP	Passerelle
net0	eth0	vmbr317	Oui		BC:24:11:4A:…	192.168.4.253/24	

Puis, il faut installer le paquet afin de commencer la configuration du serveur:

DHCP:~# apt install isc-dhcp-server

Configuration du service DHCP :

Pour la globalité de la configuration, les modifcations seront à apporter au niveau du fichier :

DHCP:~# nano /etc/dhcp/dhcpd.conf

La prémière démarche consiste à définir le srveur DHCP comme ‘générale. Pour cela, il s’uffit de décommenté la ligne « authoritative; »

```
  GNU nano 7.2                    /etc/dhcp/dhcpd.conf
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
#log-facility local7;

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

#subnet 10.152.187.0 netmask 255.255.255.0 {
#}

# This is a very basic subnet declaration.

#subnet 10.254.239.0 netmask 255.255.255.224 {
#  range 10.254.239.10 10.254.239.20;
#  option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
#}

# This declaration allows BOOTP clients to get dynamic addresses,

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line
```

Puis, décommenter toutes les lignes suivantes pour obtenir les options générales ;

```
  GNU nano 7.2                    /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "dortmund.cub.fr";
option domain-name-servers 8.8.8.8;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
                              [ Read 107 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute    ^C Location
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify    ^/ Go To Line
```

Enfin, il faut modifer les options présentes dans le **subnet** (celui-ci définit l'adresse de réseau avec le masque de sous-réseau « netmask ».

(**range** définit la **plage IP à utiliser** pour les hôtes ;

domain-name-servers et **domain-name** **définissent** les informations du serveur de nom et de nom de domaine ;

option routers **définit** l'adresse de passerelle ;

option broadcast-address **définit** l'adresse de broadcast;)

```
  GNU nano 7.2                                        /etc/dhcp/dhcpd.conf
# A slightly different configuration for an internal subnet.
 subnet 192.168.4.0 netmask 255.255.255.0 {
  range 192.168.4.1 192.168.4.252;
  option domain-name-servers 8.8.8.8;
  option domain-name-servers 172.16.4.10;
  option domain-name "dortmund.cub.fr";
  option routers 192.168.4.254;
  option broadcast-address 192.168.4.255;
#  default-lease-time 600;
#  max-lease-time 7200;
}

# Hosts which require special configuration options can be listed in
# host statements.   If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific information
# will still come from the host declaration.

#host passacaglia {
#  hardware ethernet 0:0:c0:5d:bd:95;
#  filename "vmunix.passacaglia";
#  server-name "toccata.example.com";
#}

# Fixed IP addresses can also be specified for hosts.   These addresses
# should not also be listed as being available for dynamic assignment.
# Hosts for which fixed IP addresses have been specified can boot using
# BOOTP or DHCP.   Hosts for which no fixed address is specified can only
# be booted with DHCP, unless there is an address range on the subnet
# to which a BOOTP client is connected which has the dynamic-bootp flag
```

Ensuite il faut indiquer l'interface (carte réseau) sur la qu'elle le serveur DHCP écoute et vérifier que l'interface indiquée est bien la bonne interface:

```
  GNU nano 7.2                    /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth0"
INTERFACESv6=""

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line
```

Enfin, redémarrer le serveur avec : **systemctl restart isc-dhcp-server**

TEST du service DHCP :

Afin de vérifier le bon fonctionnement du service, il faut le tester sur un client.

Pour mettre un hôte sur le réseau et tester la demande d'attribution d'une adresse IP, il faut : **ifdown eth0** et **ifup eth0**

```
root@Client:~# ifdown eth0
Killed old client process
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/bc:24:11:fb:de:a3
Sending on   LPF/eth0/bc:24:11:fb:de:a3
Sending on   Socket/fallback
DHCPRELEASE of 192.168.4.1 on eth0 to 192.168.4.253 port 67
```

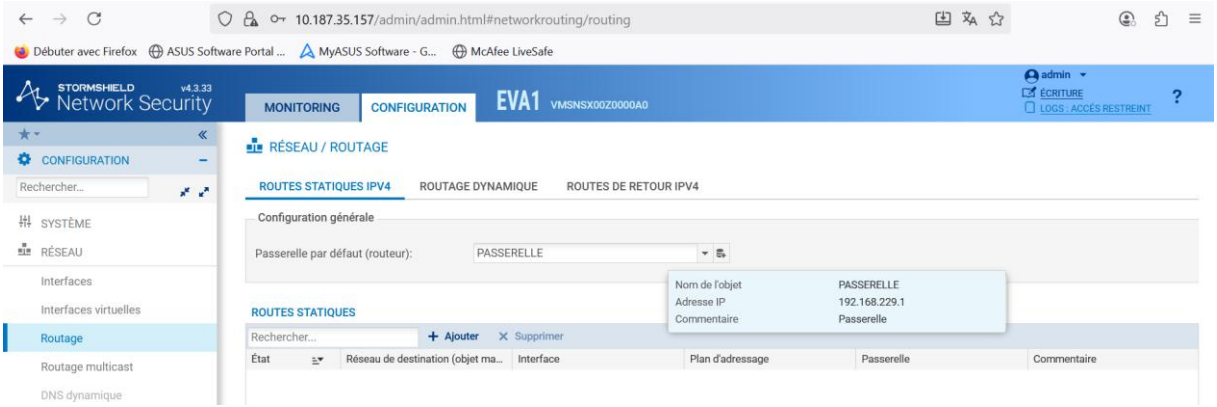
```
root@Client:~# ifup eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/bc:24:11:fb:de:a3
Sending on   LPF/eth0/bc:24:11:fb:de:a3
Sending on   Socket/fallback
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 3
DHCPOFFER of 192.168.4.1 from 192.168.4.253
DHCPREQUEST for 192.168.4.1 on eth0 to 255.255.255.255 port 67
DHCPACK of 192.168.4.1 from 192.168.4.253
bound to 192.168.4.1 -- renewal in 247 seconds.
root@Client:~#
```

Puis, avec un **ip a**, nous pouvons constater l'adresse du client obtenue.

```
root@Client:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0@if939: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group defa
ult qlen 1000
    link/ether bc:24:11:fb:de:a3 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 192.168.4.1/24 brd 192.168.4.255 scope global dynamic eth0
       valid_lft 580sec preferred_lft 580sec
    inet6 fe80::be24:11ff:fefb:dea3/64 scope link
       valid_lft forever preferred_lft forever
root@Client:~#
```

Attribution des adresses au port

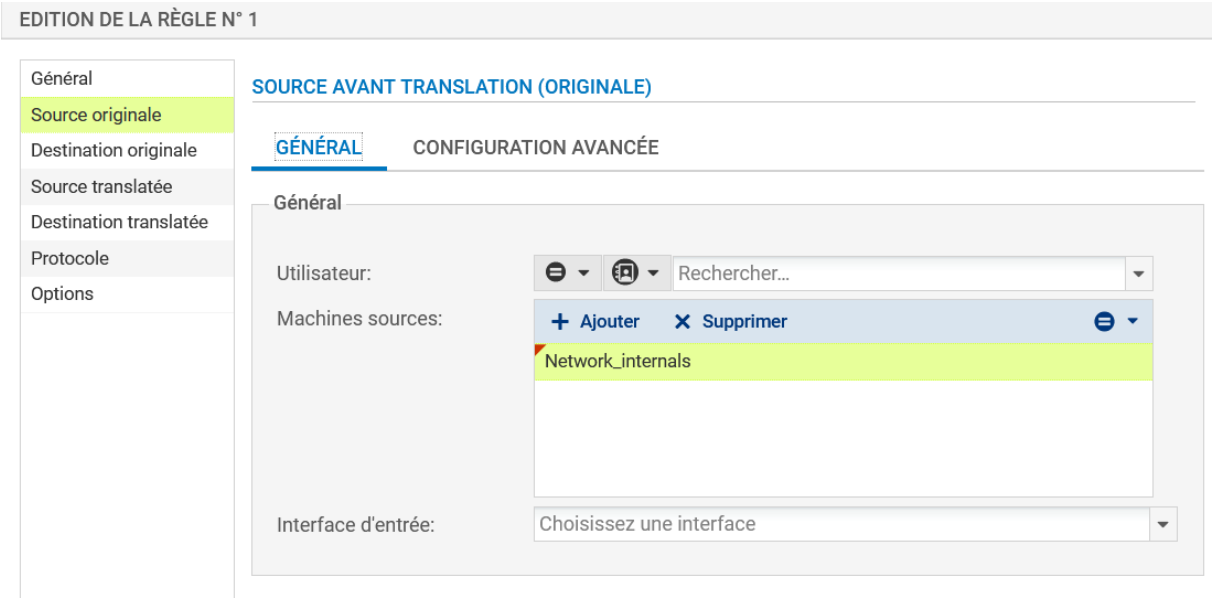


Passerelle :

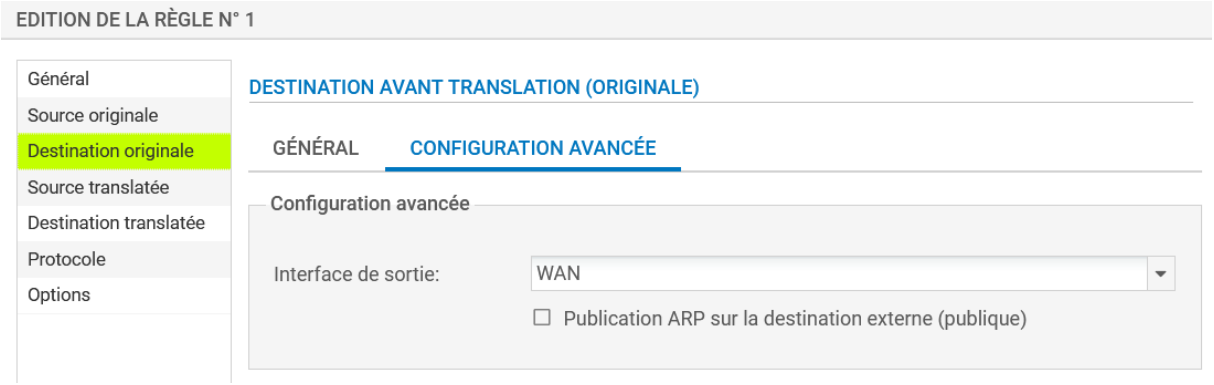
Interface	Port	Type	État	Adresse IPv4
WAN	1	Ethernet, 10 Gbit/s		192.168.229.15/24
LAN	2	Ethernet, 10 Gbit/s		192.168.4.254/24
SERVEURS	3	Ethernet, 10 Gbit/s		172.16.24.254/24
DMZ	4	Ethernet, 10 Gbit/s		172.16.4.254/24
WIFI	5	Ethernet, 10 Gbit/s		10.187.35.157/24 (DHCP)

NAT :

Créer une règle + masquerading



Choisir comme source originale « Network_internals »



Choisir WAN

EDITION DE LA RÈGLE N° 1

- Général
- Source originale
- Destination originale
- Source translatée
- Destination translatée
- Protocole
- Options

SOURCE APRÈS TRANSLATION

GÉNÉRAL | CONFIGURATION AVANCÉE

Général

Machine source translatée: Firewall_WAN

Port source translaté: ephemeral_fw

choisir aléatoirement le port source translaté

Choisir Firewall_WAN + cocher « choisir aléatoirement le port translaté »

- Général
- Source originale
- Destination originale
- Source translatée
- Destination translatée
- Protocole
- Options

PROTOCOLE

Protocole

Type de protocole: Détection automatique du protocole (par défaut)

Protocole applicatif: Basé sur les ports par défaut ou le contenu

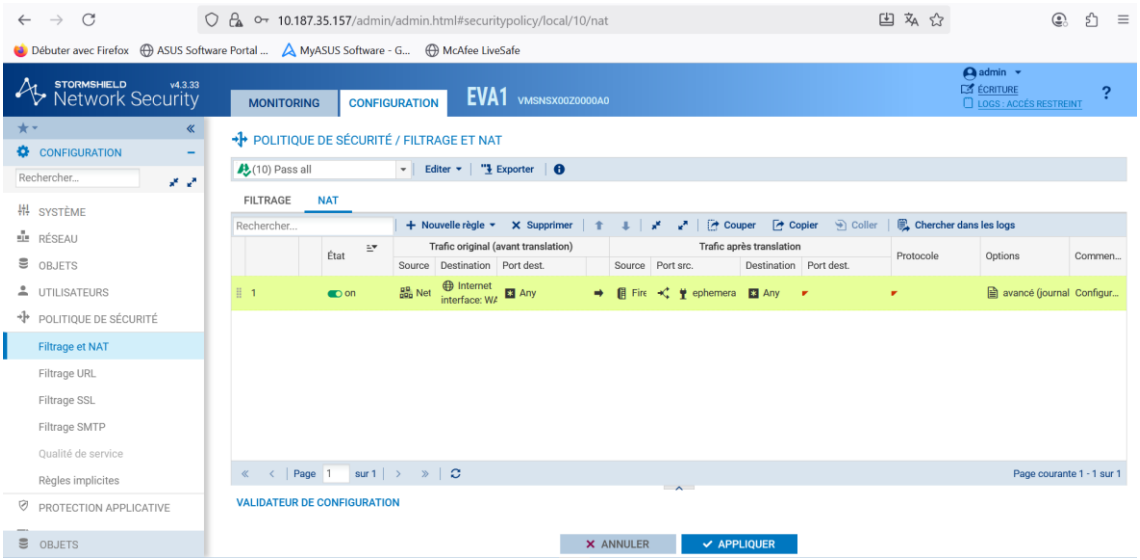
Protocole IP: Tous

- Général
- Source originale
- Destination originale
- Source translatée
- Destination translatée
- Protocole
- Options

OPTIONS

Niveau de trace: tracer

☐ NAT dans le tunnel IPsec (avant chiffrement, après déchiffrement)



CREATION DES REGLES :

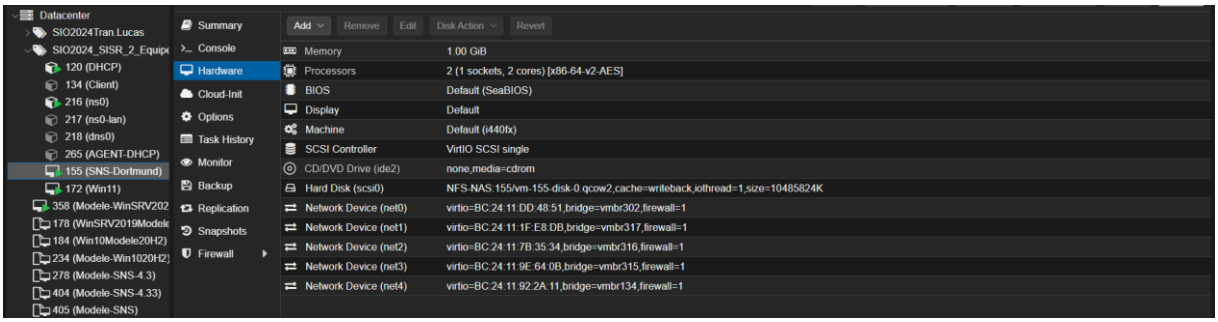
SECTION 1 :

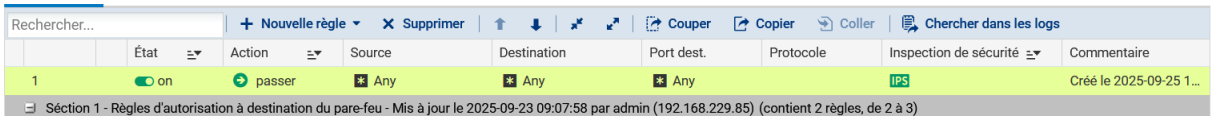
Séction 1 - Règles d'autorisation à destination du pare-feu (contient 3 règles, de 1 à 3)							
1	on	passer	Admin	Firewall_all	http https	IPS	Admin WEB/DNS
2	on	passer	Admin	Firewall_all	ssh	IPS	Admin HTTPS/SSH

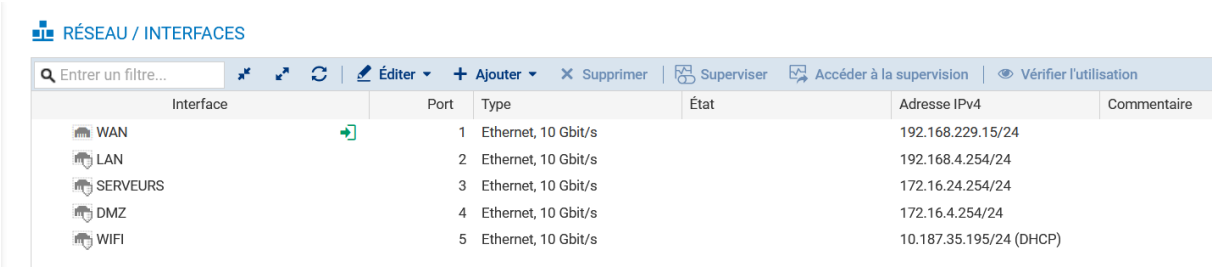
SECTION 4 :

Séction 4 - Règles d'autorisation des flux métiers (contient 3 règles, de 4 à 6)								
4	off	bloquer	Any	Any	Any		IPS	Créé le 2025-09-23 0...
5	on	passer	Network_internals	dns1.google.com	dns		IPS	Créé le 2025-09-23 0...
6	on	passer	Network_internals	Internet	Any	icmp	IPS	Créé le 2025-09-23 0...

Mise en place du DNS:







Port DNS 53

```
  GNU nano 7.2                          /etc/resolv.conf
# --- BEGIN PVE ---
search dortmund.cub.fr
nameserver 192.168.4.10
# --- END PVE ---
```

```
  GNU nano 7.2                          /etc/bind/named.conf.options

        //========================================================================
        // If BIND logs error messages about the root key being expired,
        // you will need to update your keys.  See https://www.isc.org/bind-keys
        //========================================================================

        allow-query {
          192.168.4.0/24;
          172.16.4.0/24;
        };
        dnssec-validation auto;
        #recursion no;
        version none;

        listen-on-v6 { none; };
};
```

```
  GNU nano 7.2                          /etc/bind/named.conf.local
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "dortmund.cub.fr" {
    type master;
    file "/etc/bind/db.dortmund.cub.fr";
    allow-update { 172.16.24.0/24; 192.168.4.0/24; };
};

zone "_msdcs.dortmund.cub.fr" IN {
        type master;
        file "/etc/bind/_msdcs.dortmund.cub.fr";
        allow-update { 172.16.24.0/24; 192.168.4.0/24;};
};
```

```
  GNU nano 7.2                                  db.dortmund.cub.fr
; DO NOT EDIT THIS FILE - it is used for multiple zones.
; Instead, copy it, edit named.conf, and use that copy.
;
$TTL    86400
dortmund.cub.fr.        IN      SOA     ns0.dortmund.cub.fr. root.dortmund.cub.fr. (
                              1         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                          86400 )       ; Negative Cache TTL
;
dortmund.cub.fr.        IN      NS      ns0.dortmund.cub.fr.
@               IN      A       172.16.4.10
ns0             IN      A       172.16.4.10
www             IN      A       192.168.4.35
dhcp            IN      A       192.168.4.253
```

DNS0 recursif

```
  GNU nano 7.2                          /etc/unbound/unbound.conf
interface: 192.168.4.10
interface: 127.0.0.1

access-control: 192.168.4.0/24 allow
access-control: 127.0.0.0/8 allow
access-control: 0.0.0.0/0 refuse

hide-version: yes
hide-identity: yes

do-ip4: yes

logfile: /var/log/unbound/unbound.log
verbosity: 2

private-domain: cub.fr
# La ligne suivante    ajouter car le domaine est local et il ne passe pas r  ellement par la r>
```

```
  GNU nano 7.2                                  /etc/unbound/unbound.conf
stub-zone:
name: "cub.fr."
stub-addr: 192.168.229.1

# r  cursivite pour les domaines sur Internet
forward-zone:
name: "."
forward-addr: 8.8.8.8
forward-addr: 8.8.4.4

stub-zone:
name: "dortmund.cub.fr."
stub-addr: 172.16.4.10

stub-zone:
name: "lan.dortmund.cub.fr."
stub-addr: 172.16.4.20
```